

Universal Coset Field Theory: An Albert–EIII Swampland Criterion for Heterotic Vacua

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Abstract

We develop Universal Coset Field Theory (UCFT), a four-dimensional quantum field theory built from a single quantum-mechanical datum—the state space of the exceptional Jordan (Albert) algebra $\mathbf{J}_3(\mathbb{O})$, with reduced structure group $\mathbf{E}_6$ —and use it to formulate a new, sharp swampland criterion for heterotic string vacua. An $\mathbf{E}_6$ -invariant coercive potential attains its minimum precisely on Cartan’s Hermitian symmetric space EIII, $\mathbf{E}_6/(\mathbf{SO}(10) \times \mathbf{U}(1))$, whose **32** real tangent directions form a single *irreducible* $\mathbf{SO}(10) \times \mathbf{U}(1)$ multiplet with complexification $\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ and carry a unique, positive-definite invariant metric. A spacetime-filling configuration of these Goldstone fields immerses a four-manifold into EIII with nondegenerate Riemannian first fundamental form; Lorentzian spacetime follows by Osterwalder–Schrader reconstruction, and one-loop fluctuations induce an Einstein–Hilbert term in the sense of Sakharov, with no fundamental graviton. We then show that this coset sector is the gauge-unfixed Higgs orbit of an $\mathbf{E}_8 \times \mathbf{E}_8$ heterotic $\mathbf{E}_6$ compactification: the visible two-derivative bosonic action and $\mathbf{SO}(10) \times \mathbf{U}(1)_\psi$ current algebra match those of UCFT, a gravity-from-field-theory correspondence realized concretely by a $\mathbb{Z}_{12}$ asymmetric orbifold with three generations and an adjoint Higgs. Our main result is the *Albert–EIII swampland theorem*: within the Albert-selected heterotic class, a four-dimensional effective theory lies in the landscape only if its gauge-unfixed Higgs/moduli sector is the EIII coset with the stated tangent representation, invariant metric, gauge data, anomaly and Green–Schwarz matching, and worldsheet consistency; every other Albert-selected effective theory lies in the Albert–EIII swampland, admitting no heterotic UV completion realizing the UCFT correspondence. This is a relative, class-restricted statement, not a universal one, and it makes the $\mathbb{Z}_{12}$ model an explicit leading landscape candidate. A functional renormalization-group analysis exhibits a unique asymptotically safe

fixed point with grand-unified scale $M_{\text{GUT}} \simeq 2 \times 10^{16} \text{ GeV}$. We separate what is fixed by representation theory from what requires further dynamical input, and delineate the open problems.

Keywords: swampland, exceptional Jordan algebra, E_6 , heterotic string, $\mathbb{Z}_{12}$ orbifold, coset field theory, EIII, induced gravity, emergent spacetime, asymptotic safety, grand unification

1 Introduction

Exceptional algebraic structures recur throughout high-energy physics. The exceptional Jordan (Albert) algebra $J_3(\mathbb{O})$ of 3×3 Hermitian octonionic matrices has automorphism group F_4 and reduced structure group $E_6 = \text{Str}_0(J_3(\mathbb{O}))$, the invariance group of its cubic norm [1–3]. In turn E_6 contains $\text{SO}(10) \times \text{U}(1)$, under which the fundamental **27** branches as

$$\mathbf{27} = \mathbf{16}_{+1} \oplus \mathbf{10}_{-2} \oplus \mathbf{1}_{+4},$$

the **16** of $\text{Spin}(10)$ carrying one Standard-Model family together with a right-handed neutrino [4]. Formally real Jordan algebras were introduced precisely as the algebra of observables of a quantum system; $J_3(\mathbb{O})$ is the unique exceptional member of their classification. This gives a sharp conceptual starting point: take a quantum state space as the only datum and ask how much of four-dimensional physics its internal symmetry fixes.

The construction rests on two axioms, stated in Section 2.1. The first posits a maximally symmetric initial state in $J_3(\mathbb{O})$; the second posits the usual quantum-mechanical kinematics of a universal wavefunction with unitary evolution. From these we build a field theory, not by adding structure, but by following the consequences of symmetry breaking on the Jordan state space. An E_6 -invariant polynomial potential, built from the quadratic and cubic invariants Q and N , is coercive and is minimized exactly on the orbit

$$E_6 \cdot X_0 \cong \frac{E_6}{\text{SO}(10) \times \text{U}(1)},$$

the compact Hermitian symmetric space EIII [5], of real dimension $78 - 46 = 32$. The 32 Goldstone directions form a single irreducible multiplet of $H = \text{SO}(10) \times \text{U}(1)$, whose complexification is $\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$; the invariant metric is therefore unique up to scale and, since H is compact, positive definite.

The central claim of the paper is that spacetime and gravity are not postulated but emerge from this coset sector. A spacetime-filling configuration of the Goldstone fields immerses a four-manifold into EIII; its first fundamental form is a genuine, nondegenerate Riemannian metric, so the four-dimensional geometry is induced by the field configuration rather than fixed by hand. Because the functional integral is defined in Euclidean signature—as required for the Osterwalder–Schrader reconstruction invoked in Axiom 2—physical Lorentzian spacetime is recovered by continuation. Quantizing the 32 scalars, the gauge fields, and a chiral **16**, one-loop fluctuations then induce

an Einstein–Hilbert term in the sense of Sakharov [6, 7]; the metric propagates only through this induced action, and there is no fundamental graviton.

Two further results complete the picture. First, a functional renormalization-group analysis of the coset-field action exhibits a unique ultraviolet-attractive non-Gaussian fixed point at which all higher-curvature and higher-scalar operators are irrelevant; within the truncation the theory is asymptotically safe, and downward running of the gauge couplings fixes a grand-unified scale $M_{\text{GUT}} \simeq 2 \times 10^{16} \text{ GeV}$. Second, the E_6 branchings supply the grand-unified matter skeleton and its anomaly conditions. We are careful to separate theorem from construction from conjecture. What is proved: the coercive potential and its EIII minimum, the irreducibility and unique positive metric of the 32-scalar sector, the emergent Riemannian geometry, the induced Einstein–Hilbert term, and the fixed-point structure of the truncated flow. What is supplied by additional dynamical input, and left open: the complete low-energy spectrum, family replication, the cosmological constant, and a full ultraviolet completion.

The paper is organized as follows. [Section 2](#) states the axioms, recalls the Albert algebra and its E_6 symmetry, and constructs the symmetry-breaking potential and its vacuum manifold. [Section 3](#) derives the emergent four-dimensional geometry and the low-energy field content. [Section 4](#) defines the coset σ -model, quantizes it, and derives the induced Einstein–Hilbert term and the coset mass spectrum. [Section 5](#) presents the renormalization-group flow, the non-Gaussian fixed point, and the grand-unified matching scale. Proofs and technical computations are collected in the appendices.

2 Axioms, the Albert Algebra, and Spontaneous Symmetry Breaking

In this section, we introduce our two fundamental axioms, recall the definition and key properties of the Albert algebra $J_3(\mathbb{O})$ and its E_6 symmetry (with proofs in [Appendix A](#)), exhibit a purely algebraic polynomial potential whose minima break $E_6 \rightarrow \text{SO}(10) \times \text{U}(1)$, and count the resulting 32 Goldstone directions.

2.1 Axioms

Axiom 1 (Maximally symmetric vacuum) *The universe begins in a unique state $\Phi \in J_3(\mathbb{O})$ that is fixed by every element of its structure group.*

Axiom 2 (Universal wavefunction) *Physical states $\psi \in \mathcal{H}$ evolve through a one-parameter family $\{U(t)\}_{t \in \mathbb{R}}$ of unitary operators satisfying $U(0) = \text{Id}$, $U(t+s) = U(t)U(s)$, and strong continuity.*

Remark 1 [Axiom 2](#) provides the kinematical Hilbert space and the one-parameter family of unitary operators. Whenever we later write a functional integral, we treat it as a formal generator of connected Green functions; its constructive definition follows from the Osterwalder-Schrader reconstruction theorem applied to the Euclidean correlation functions [8].

No additional axioms are introduced; all further constructions follow as definitions or theorems derived from these two axioms.

2.2 The Albert Algebra

Proofs for this subsection are provided in [Appendix A](#).

Definition 1 (Jordan algebra [1]) Let F be a field with $\text{char } F \neq 2$. A *Jordan algebra* is a commutative algebra $(A, \circ)$ satisfying

$$a \circ (b \circ a^2) = (a \circ b) \circ a^2, \quad a, b \in A.$$

Definition 2 (Albert algebra [3, 9]) The *Albert algebra* is

$$J_3(\mathbb{O}) = \{X \in M_3(\mathbb{O}) : X = X^\dagger\}, \quad X \circ Y = \frac{1}{2}(XY + YX).$$

Theorem 3 The Albert algebra is a 27-dimensional real Jordan algebra.

Theorem 4 (Quadratic and cubic invariants) $J_3(\mathbb{O})$ admits

$$Q(X) = \text{Tr}(X^2), \quad N(X) = \det_{\mathbb{O}}(X),$$

both preserved by its structure group.

Theorem 5 (Structure group [2]) The structure group of $J_3(\mathbb{O})$ is the simply connected exceptional Lie group E_6 .

Theorem 6 (Orbit classification) For $X, Y \in J_3(\mathbb{O})$,

$$(Q(X), N(X)) = (Q(Y), N(Y))$$

if and only if there exists $g \in E_6$ such that $g \cdot X = Y$.

Corollary 7 (Existence of an $SO(10) \times U(1)$ stabiliser) There exists $X_0 \in J_3(\mathbb{O})$ with $\text{Stab}_{E_6}(X_0) \cong SO(10) \times U(1)$.

2.3 Polynomial Symmetry-Breaking Potential

Set

$$I_2(X) = Q(X), \quad I_3(X) = N(X), \quad c_2 = I_2(X_0), \quad c_3 = I_3(X_0).$$

In particular, each simultaneous level set $\{I_2 = c_2, I_3 = c_3\}$ is exactly one E_6 -orbit, so by choosing $c_2 = I_2(X_0)$ and $c_3 = I_3(X_0)$, we identify the zero locus of $P(X)$ with $E_6 \cdot X_0$. In addition, define

$$\begin{aligned} P(X) &= [I_2(X) - c_2]^2 + [I_3(X) - c_3]^2, \\ V(X) &= P(X) + \alpha [I_2(X)]^k, \end{aligned} \tag{1}$$

with $\alpha > 0$ and even $k \geq 2$.

Lemma 8 $P(X) = 0$ if and only if $X \in E_6 \cdot X_0$.

Proof By invariance of I_2, I_3 under E_6 , $P(g \cdot X_0) = 0$. Conversely, if $P(X) = 0$ then $(I_2, I_3) = (c_2, c_3)$ and [Theorem 6](#) gives $X \in E_6 \cdot X_0$. $\square$

Theorem 9 (Existence of symmetry breaking) *The polynomial $V(X)$ in [\(1\)](#) is E_6 -invariant, coercive, and attains its absolute minimum precisely on the orbit*

$$E_6 \cdot X_0 \cong \frac{E_6}{\text{SO}(10) \times \text{U}(1)}.$$

Proof Invariance and coercivity are immediate (the even-degree term αI_2^k dominates as $\|X\| \rightarrow \infty$). By [Lemma 8](#), $P \geq 0$ with $P = 0$ exactly on $E_6 \cdot X_0$, where $V = \alpha c_2^k$; elsewhere $V > \alpha c_2^k$. $\square$

2.4 Dimension and Vacuum Manifold

Theorem 10 (Dimension count)

$$\dim E_6 = 78, \quad \dim(\text{SO}(10) \times \text{U}(1)) = 46, \quad \dim\left(\frac{E_6}{\text{SO}(10) \times \text{U}(1)}\right) = 32.$$

Proof The dimension of E_6 is tabulated in [\[4\]](#). The classical formula $\dim \text{SO}(2n) = n(2n - 1)$ gives $\dim \text{SO}(10) = 45$; adding the one-dimensional $\text{U}(1)$ yields 46. The difference is 32. $\square$

Corollary 11 (Vacuum manifold) *Spontaneous breaking $E_6 \rightarrow \text{SO}(10) \times \text{U}(1)$ yields 32 Goldstone bosons parameterizing the vacuum manifold*

$$\mathcal{M} = \frac{E_6}{\text{SO}(10) \times \text{U}(1)}.$$

Remark 2 Throughout this paper, we denote by $\mathcal{M}$ the coset manifold $E_6/(\text{SO}(10) \times \text{U}(1))$.

3 Emergent Spacetime and Field Content

The vacuum manifold $\mathcal{M}$ is 32-dimensional, but a spacetime-filling field configuration immerses a four-dimensional world manifold into it. In this section, we show how a frame and a nondegenerate Riemannian metric arise as the first fundamental form of that immersion, how physical Lorentzian spacetime is recovered by continuation, and how the transverse directions combine with $\text{SO}(10) \times \text{U}(1)$ gauge connections and fermionic coherent states to furnish the matter content of UCFT.

3.1 Emergent Spacetime

Choose a local coordinate patch

$$\{\phi^\alpha(x)\}_{\alpha=0}^{31}, \quad \phi^\alpha(x) : \mathbb{R}^4 \rightarrow \mathcal{M},$$

where $\{x^\mu\}_{\mu=0}^3$ are smooth parameters on the world manifold. Write $\theta = g^{-1}dg$ for the left-invariant Maurer-Cartan one-form on E_6 and let θ_α^a denote its components along the broken generators $\mathfrak{e}_6 \setminus (\mathfrak{so}(10) \oplus \mathfrak{u}(1))$.

Remark 3 Throughout this section, the symbols x^μ serve solely as four real parameters that label the field configuration $\phi^\alpha(x)$. Their physical interpretation as spacetime coordinates is justified only after [Theorem 13](#), where the pull-back metric is shown to be non-degenerate, together with the continuation to Lorentzian signature discussed there.

Let $\mathfrak{g}_{\alpha\beta} = f^2(-K_{\alpha\beta})$ denote the unique positive-definite E_6 -invariant metric on the broken subspace $\mathfrak{m}$ ([Theorem 15](#)), where $K_{\alpha\beta}$ is the restriction of the Killing form; on the compact form $-K|_{\mathfrak{m}}$ is positive definite.

Lemma 12 (Rank-four embedding) *Let the pull-back $g_{\mu\nu} = \mathfrak{g}_{\alpha\beta} \partial_\mu \phi^\alpha \partial_\nu \phi^\beta$ be non-degenerate at x . Then the differential $d\phi_x : T_x \mathbb{R}^4 \rightarrow \mathfrak{m}$ has rank four, and its image $\mathfrak{p}_0(x) = \Im d\phi_x$ is a four-dimensional subspace of $\mathfrak{m}$; the transverse 28 Goldstone directions span its $\mathfrak{g}$ -orthogonal complement $\mathfrak{p}_m(x)$.*

Proof See [Appendix B.1](#). Since $\dim T_x \mathbb{R}^4 = 4$, one has $\text{rank } d\phi_x \leq 4$, and because $\mathfrak{g}$ is positive definite $\text{rank } g_{\mu\nu} = \text{rank } d\phi_x$. Non-degeneracy of $g_{\mu\nu}$ therefore forces $\text{rank } d\phi_x = 4$, i.e. $d\phi_x$ is injective. $\square$

Choosing a $\mathfrak{g}$ -orthonormal basis of $\mathfrak{p}_0(x)$ and pulling back the Maurer-Cartan form gives the tetrad

$$e_\mu^a(x) = \theta_\alpha^a(\phi(x)) \partial_\mu \phi^\alpha(x), \quad a = 0, \dots, 3, \quad (2)$$

which by [Lemma 12](#) is invertible for a non-degenerate configuration. The induced (first fundamental) metric is

$$g_{\mu\nu}(x) = \delta_{ab} e_\mu^a(x) e_\nu^b(x), \quad (3)$$

where δ_{ab} is the restriction of the positive-definite $\mathfrak{g}$ to the orthonormal frame $\mathfrak{p}_0(x)$.

Theorem 13 (Emergent Riemannian geometry) *For an open dense set of spacetime-filling configurations $\phi : \mathbb{R}^4 \rightarrow \mathcal{M}$, the pull-back (3) is a smooth, non-degenerate, positive-definite (Riemannian) metric on $\mathbb{R}^4$. It is the first fundamental form of the immersion ϕ into the Hermitian symmetric space $\mathcal{M}$, and no metric is postulated on $\mathbb{R}^4$ independently of ϕ .*

Proof See [Appendix B.2](#). Non-degeneracy and rank four are [Lemma 12](#); positive-definiteness is inherited from that of $\mathfrak{g}$ ([Theorem 15](#)), since $g_{\mu\nu} v^\mu v^\nu = \mathfrak{g}(d\phi_x v, d\phi_x v) \geq 0$ with equality iff $v = 0$. $\square$

The tetrad e_μ^a is thereby the fundamental gravitational field, and its dynamics follow from the σ -model action of [Section 4](#).

Remark 4 (Recovery of Lorentzian spacetime) The coset metric $\mathfrak{g}$ is positive definite, so the induced geometry of [Theorem 13](#) is necessarily Riemannian: a pull-back of a definite form cannot have indefinite signature. This is the appropriate object, because the functional integral is defined in Euclidean signature ([Axiom 2](#) and [Appendix C.2](#)). Physical Lorentzian spacetime is recovered from the Euclidean correlation functions by Osterwalder–Schrader reconstruction [8], equivalently by Wick rotation of the reconstructed frame, exactly as in Euclidean quantum gravity.

Remark 5 (Dynamical signature: outlook) An alternative, more ambitious route keeps the theory in Lorentzian signature from the outset by installing the signature as an order parameter: a distinguished clock-sector gradient condensate $\langle e_\mu^a \rangle$ that spontaneously breaks the internal frame rotations $\mathrm{SO}(4) \rightarrow \mathrm{SO}(1, 3)$. This requires an explicit signature-selecting term beyond the definite coset metric and is not needed for the results proved here; we record it as a direction for future work ([Section 4](#) uses only the Euclidean construction above).

Remark 6 (Emergent spacetime and rigidity) The image of $\mathbb{R}^4$ inside $\mathcal{M}$ is the physical spacetime manifold, and its metric (3) emerges from the field configuration rather than being put in by hand. The four frame directions $\mathfrak{p}_0(x)$ are *not* an H -invariant subspace: by [Lemma 14](#) the isotropy representation on $\mathfrak{m}$ is irreducible, so $\mathfrak{m}$ admits no H -invariant four-plane. The splitting $\mathfrak{m} = \mathfrak{p}_0 \oplus \mathfrak{p}_m$ is therefore selected *spontaneously* by the configuration $\phi(x)$, in the same sense that a vacuum expectation value selects a direction; there is no contradiction between the emergence of a preferred frame and the H -irreducibility of the coset scalars.

3.2 Low-Energy Field Content

With the vierbein $e_\mu^a(x)$ invertible and [Theorem 13](#) establishing a non-degenerate induced metric (Lorentzian after continuation, 4), $\mathbb{R}^4$ acquires the interpretation of physical spacetime. We now catalogue the dynamical fields that live on it and transform under the unbroken group $H = \mathrm{SO}(10) \times \mathrm{U}(1)$.

Introduce the connection one-form

$$A_\mu(x) = A_\mu^A(x) T_A \in \mathfrak{h}, \quad (4)$$

which transforms by the affine shift $A_\mu \mapsto h A_\mu h^{-1} + h \partial_\mu h^{-1}$ for $h(x) \in H$. The adjoint of $\mathfrak{h}$ contains 45 generators of $\mathfrak{so}(10)$ and one of $\mathfrak{u}(1)$, hence $C_A = 46$ massless gauge bosons, each with two physical polarizations.

The theory contains a single left-handed Weyl multiplet in the chiral spinor $\mathbf{16}_{+1}$ of $\mathrm{Spin}(10)$; all fermions are sections of the bundle $S \otimes \mathcal{V}_{\mathbf{16}}$, where S is the Weyl spin bundle determined by the emergent vierbein [10]. Counting two real components per Weyl fermion gives $n_F = 16$ real degrees of freedom.

The 32 coordinates of $\mathcal{M}$ form a single real-irreducible multiplet of $H = \mathrm{SO}(10) \times \mathrm{U}(1)$ ([Lemma 14](#)), whose complexification is $\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$. There is thus no H -invariant $4 + 28$ decomposition; the four frame directions $\mathfrak{p}_0$ singled out by [Lemma 12](#) are

selected *spontaneously* by the spacetime-filling configuration (6). Relative to a given background, the four longitudinal directions build the tetrad, while the 28 transverse fluctuations $\mathbf{p}_m$ acquire a mass of order $f\mu$ (Theorem 18). All $n_S = 32$ real scalars enter the functional renormalization-group flow of Section 5.

Table 1 UCFT’s low-energy field content.

Sector	H representation	Spin	Real d.o.f.	Multiplicity n
Coset scalars	$\mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$	0	32	n_S
Chiral fermions	$\mathbf{16}_{+1}$	$\frac{1}{2}$	32	n_F
Gauge bosons	adj_H	1	92	C_A

Remark 7 On a fixed non-degenerate background the tetrad e_μ^a maps the world-manifold index μ to the four-dimensional frame $\mathbf{p}_0(x)$ selected by that configuration, and so acts as a *soldering form* between the emergent geometry and the coset. This soldering is a property of the configuration, not an H -invariant identification (6); it ensures that all matter fields couple consistently to the induced frame.

Collecting e_μ^a , A_μ , the chiral fermion ψ , and the coset scalars ϕ^α , we have now assembled the complete low-energy spectrum of UCFT.

4 Coset σ -Model, Quantization, and Induced Gravity

In this section, we construct the σ -model whose target space is the coset $\mathcal{M}$, show that its path-integral quantization defines a four-dimensional quantum field theory, and prove that one-loop matter fluctuations induce an Einstein-Hilbert term with a Planck scale set by the σ -model decay constant.

4.1 Maurer-Cartan Form and Invariant Metric

Let $g(x) \in E_6$ be a local coset representative. The left-invariant Maurer-Cartan form is

$$\theta_\mu(x) = g^{-1}(x) \partial_\mu g(x) \in \mathfrak{e}_6. \quad (5)$$

Decomposing θ_μ into unbroken $\mathfrak{h} = \mathfrak{so}(10) \oplus \mathfrak{u}(1)$ and broken $\mathfrak{m}$ components gives

$$\theta_\mu = A_\mu^i H_i + e_\mu^\alpha X_\alpha,$$

with $H_i \in \mathfrak{h}$, $X_\alpha \in \mathfrak{m}$, and coefficient functions $A_\mu^i(x)$ and $e_\mu^\alpha(x)$. α, β run over the 32 broken generators while i, j label the 45 generators of $\mathfrak{h}$.

Lemma 14 *The isotropy representation of $H = \text{SO}(10) \times \text{U}(1)$ on the broken subspace $\mathfrak{m}$ is irreducible.*

Proof Inspecting the E_6 root diagram [5], one finds that the 32 broken generators form a single weight orbit under H ; hence no proper non-trivial H -invariant subspace of $\mathfrak{m}$ exists. $\square$

Theorem 15 *Up to an overall positive constant f^2 , there exists a unique E_6 -invariant symmetric bilinear form $\mathfrak{g}_{\alpha\beta}$ on $\mathfrak{m}$, namely $\mathfrak{g}_{\alpha\beta} = f^2(-K_{\alpha\beta})$, where $K_{\alpha\beta}$ is the restriction to $\mathfrak{m}$ of the Killing form on $\mathfrak{e}_6$. On the compact real form of E_6 the Killing form is negative definite, so $\mathfrak{g}_{\alpha\beta}$ is positive definite.*

Proof See Appendix C.1 for a short Schur-lemma argument. $\square$

4.2 Classical σ -Model Action

By the coset construction of nonlinear realizations [11], the broken generators X_α furnish local coordinates $\phi^\alpha(x)$ on the vacuum manifold $\mathcal{M}$. The target-space metric $\mathfrak{g}_{\alpha\beta}$ is the unique E_6 -invariant metric (Theorem 15), and the spacetime metric $g_{\mu\nu}$ emerges from the vierbein pull-back construction of Section 3.1. As in Section 3.2, we introduce the $\mathfrak{h}$ -valued gauge connection $A_\mu = A_\mu^i T_i$, and define

$$D_\mu \phi^\alpha = \partial_\mu \phi^\alpha + A_\mu^i (t_i)^\alpha{}_\beta \phi^\beta, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu],$$

where $(t_i)^\alpha{}_\beta$ are the representation matrices of $\mathfrak{h}$ on the coset coordinates.

By standard effective-field-theory arguments [11, 12], the low-energy action is given by the most general derivative expansion consistent with the symmetries. Truncating at two spacetime derivatives and imposing local H - and Lorentz invariance uniquely fixes its form. Introducing a single left-handed Weyl fermion $\psi(x)$ in the $\mathbf{16}_{+1}$ of $\text{Spin}(10)$, one obtains

$$S[\phi, A, \psi] = \int d^4x \sqrt{-g} \left[\frac{1}{2} g^{\mu\nu} \mathfrak{g}_{\alpha\beta} D_\mu \phi^\alpha D_\nu \phi^\beta - \frac{1}{4} g^{-2} \text{Tr}(F_{\mu\nu} F^{\mu\nu}) + \bar{\psi} i \gamma^\mu D_\mu \psi - y \bar{\psi} \phi^\alpha T_\alpha \psi \right]. \quad (6)$$

The dimensionless couplings g and y , together with the decay constant f , depend on the renormalization scale; their flow is studied in Section 5.

Lemma 16 *For $f^2 > 0$, the kinetic term in (6) is positive definite, and hence the classical Hamiltonian is bounded below.*

Proof Since $\mathfrak{g}_{\alpha\beta}$ is positive definite on the broken subspace, $D_\mu \phi^\alpha D^\mu \phi^\beta \mathfrak{g}_{\alpha\beta} \geq 0$, and all remaining terms enter at most quadratically. $\square$

4.3 Path Integral, Induced Gravity, and Mass Spectrum

We quantize the nonlinear sigma model action (6) via the path-integral formalism [13, 14]. Gauge redundancy under local $\text{SO}(10) \times \text{U}(1)$ is fixed by the Faddeev-Popov procedure [13]; the resulting BRST symmetry is developed in Appendix C.2. The partition function is

$$Z = \int \mathcal{D}\phi \mathcal{D}A \mathcal{D}\psi \exp[-S[\phi, A, \psi]]. \quad (7)$$

Expanding around a slowly varying background $X_0 \in \mathcal{M}$ and integrating out the quadratic fluctuations by Gaussian functional integration yields the one-loop effective action [15]

$$\Gamma^{(1)} = \frac{1}{2} \log \det(-\Delta_\phi) - \log \det(-\Delta_{\text{ghost}}) + \cdots, \quad (8)$$

where $-\Delta_\phi$ and $-\Delta_{\text{ghost}}$ are the second-variation operators for the physical and ghost fields, respectively. Its quadratic expansion in the fluctuation $\varphi = X - X_0 \in T_{X_0} \mathcal{M}$ reads

$$\Gamma^{(1)}[X_0 + \varphi] = \Gamma^{(1)}[X_0] + \frac{1}{2} \int d^4x \varphi^I [\text{Hess}(X_0)]_{IJ} \varphi^J + \cdots, \quad (9)$$

so that the eigenvalues of the Hessian of the effective potential govern the one-loop mass spectrum (Theorem 18). The induced Einstein-Hilbert term and higher-curvature operators then follow from the standard heat-kernel expansion, as stated in Theorem 17.

Remark 8 We evaluate all functional determinants in Euclidean signature on a four-torus $\mathbb{T}^4$ of finite volume V and later take the thermodynamic limit $V \rightarrow \infty$; the final result is regulator-independent.

Theorem 17 (Induced gravity) *One-loop quantum fluctuations induce an Einstein-Hilbert term*

$$\Gamma^{(1)} \supset \frac{1}{2} M_{\text{Pl}}^2 \int d^4x \sqrt{-g} R, \quad M_{\text{Pl}}^2 = \frac{N_s f^2}{(4\pi)^2} \log \frac{\Lambda^2}{\mu^2}, \quad (10)$$

where $N_s = 32$ is the number of Goldstone scalars, Λ the cutoff, and μ the renormalization scale.

Proof See Appendix C.3, which follows Sakharov's induced-gravity argument and employs the heat-kernel coefficients listed in Table C1. $\square$

Theorem 18 (Mass spectrum) *The one-loop effective potential is H -invariant, and $\mathfrak{m}$ is H -irreducible (Lemma 14); hence its Hessian at the vacuum is a single scalar,*

$$\text{Hess}(X_0) = f^2 \mu^2 \mathbf{1}_{32}, \quad m^2 = f^2 \mu^2 > 0.$$

On a spacetime-filling background the four longitudinal directions $\mathfrak{p}_0$ of Lemma 12 are the tetrad (gravitational) degrees of freedom; the remaining 28 transverse fluctuations $\mathfrak{p}_m$ are physical scalars of common mass $m^2 = f^2 \mu^2$.

Proof See [Appendix C.4](#). The tree potential of [Theorem 9](#) is E_6 -invariant and vanishes on the entire orbit $\mathcal{M}$, so all 32 coset scalars are exact Goldstone bosons at tree level. The gauge and Yukawa couplings of (6) respect only H , so the one-loop effective potential is H -invariant but not E_6 -invariant and lifts these pseudo-Goldstones. Its Hessian commutes with H ; since $\mathfrak{m}$ is H -irreducible ([Lemma 14](#)), Schur’s lemma forces $\text{Hess}(X_0) = m^2 \mathbf{1}_{32}$ with a single $m^2 = f^2 \mu^2 > 0$ fixed by the flow ([Section 5.2](#)). On a non-degenerate background the frame condensate $\langle \partial_\mu \phi^a \rangle = e_\mu^a$ reinterprets the four directions $\mathfrak{p}_0$ as the tetrad, promoted to the gravitational field by the induced Einstein-Hilbert term of [Theorem 17](#); the 28 transverse fluctuations remain as physical scalars, each of mass $m^2 = f^2 \mu^2$. $\square$

4.4 Higher-Curvature Operators

The coefficient a_4 in the heat-kernel expansion generates

$$\alpha R^2 + \beta R_{\mu\nu} R^{\mu\nu} + \gamma R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma}.$$

Theorem 19 *The dimensionless couplings α, β, γ scale as k^{-2} at large momentum scale k ; the corresponding operators are irrelevant at the non-Gaussian fixed point of [Section 5](#).*

Proof Since $a_4 \propto f^0$ while $M_{\text{Pl}}^2 \propto f^2$, one has $\alpha \sim a_4/M_{\text{Pl}}^2 \propto k^{-2}$; the same reasoning applies to β and γ . $\square$

4.5 Hierarchy of Physical Scales

Scale	Definition	Order of Magnitude
Goldstone decay constant	f	input
Planck mass	M_{Pl}	$\gtrsim 10^{18} \text{ GeV}$
GUT scale	M_{GUT}	$\simeq 2 \times 10^{16} \text{ GeV}$
Electroweak VEV	v	246 GeV

Remark 9 The value for M_{GUT} is derived in [Section 5.5](#).

5 Functional Renormalization-Group Flow

We now track the coset-field theory under continuous coarse-graining. After introducing the functional renormalization-group (RG) machinery and the dimensionless couplings, we compute the one- and two-loop β -functions, identify a unique ultraviolet-attractive non-Gaussian fixed point, determine the grand-unified matching scale, and prove that every higher operator is irrelevant. The 27×27 matrix governing classically marginal coset-scalar interactions is diagonalized in [Appendix E](#); all analytic two-loop coefficients are derived in [Appendix D](#) and summarized in [Table D2](#).

5.1 Setup

We work with the Wetterich equation

$$\partial_k \Gamma_k = \frac{1}{2} \text{Tr}[(\Gamma_k^{(2)} + R_k)^{-1} \partial_k R_k],$$

choosing Litim's regulator $R_k(p^2) = Z_k(k^2 - p^2)\theta(k^2 - p^2)$, which optimizes threshold integrals while preserving locality. All physical conclusions are regulator-independent. The natural dimensionless couplings are

$$\xi = \frac{k^2}{f^2}, \quad \hat{g}^2 = \frac{k^2 g^2}{f^2}, \quad \hat{y}^2 = \frac{k^2 y^2}{f^2}, \quad \kappa = \frac{k^2}{M_{\text{Pl}}^2}.$$

Projecting the flow onto the coefficients of $\sqrt{-g} R$, $\sqrt{-g} \bar{\psi} \gamma^\mu D_\mu \psi$, and analogous invariants yields the β -functions quoted below.

Remark 10 We evaluate functional determinants on a finite four-torus $\mathbb{T}^4$ and take $V_{\mathbb{T}^4} \rightarrow \infty$ at the end, removing all infrared regulators without affecting ultraviolet behavior.

5.2 One-loop β -Functions

At one loop, the running of the gauge, scalar and gravitational couplings is completely fixed by the field content listed in [Table 1](#). Working in the minimal-subtraction scheme (or, equivalently, in the functional RG, with a Litim regulator), the standard group-theoretic result for a gauge algebra $\mathfrak{g}$ with n_F Weyl fermions and n_S real scalars in the fundamental representation reads [\[16\]](#)

$$\beta_{g^{-2}} = \frac{b_0}{48\pi^2}, \quad b_0 = \frac{11}{3} C_A - \frac{4}{3} T_R n_F - \frac{1}{6} T_R n_S.$$

Within E_6 , we have $C_A = 12$, $T_R = 1$, and matter multiplicities $(n_F, n_S) = (16, 32)$, yielding $b_0 = 52/3$.

Throughout the paper, we trade the dimensionless gauge coupling $g^2(k)$ for the canonically rescaled

$$\hat{g}^2(k) = \frac{k^2}{f^2} g^2(k),$$

so that $\hat{g}^2$ has engineering dimension 2. Using $\beta_{\hat{g}^2} = 2\hat{g}^2 \beta_g / g$, we find

$$\beta_{\hat{g}^2} = -\frac{b_0}{24\pi^2} \hat{g}^4$$

The coset-field wave-function f^2 is protected by the E_6 symmetry at this order, hence $\beta_{f^2} = 0$. The dimensionful ratio $\xi(k) = k^2/f^2$ therefore runs only through its canonical dimension: $\beta_\xi = 2\xi$.

Finally, integrating out the 28 massive coset fields generates the one-loop Sakharov flow of the induced Planck mass [6, 7]:¹

$$\beta_\kappa = 2\kappa + \frac{5}{48\pi^2}\kappa^2, \quad \kappa = \frac{k^2}{M_{\text{Pl}}^2(k)}.$$

Collecting the three directions, we have

$$\beta_{\hat{g}^2} = -\frac{b_0}{24\pi^2}\hat{g}^4, \quad \beta_\xi = 2\xi, \quad \beta_\kappa = 2\kappa + \frac{5}{48\pi^2}\kappa^2. \quad (11)$$

These one-loop coefficients are the only input needed for the two-coupling truncation studied in Section 5.4.

5.3 Two-Loop β -Functions

Coset-scalar self-interactions and Yukawa diagrams enter at two loops. Writing only the leading structures,

$$\begin{aligned} \beta_{\hat{g}^2} &= -\frac{b_0}{24\pi^2}\hat{g}^4 - \frac{b_1}{(16\pi^2)^2}\hat{g}^6 + \frac{c_1}{16\pi^2}\hat{g}^4\hat{g}^2, \\ \beta_{\hat{y}^2} &= b_y\hat{y}^4 + c_2\hat{g}^2\hat{y}^2, \\ \beta_\xi &= 2\xi + A\hat{y}^2\xi + b_0\hat{g}^2\xi. \end{aligned} \quad (12)$$

The analytic coefficients and their E_6 values appear in Table D2; derivations are given in Appendix D. The 27×27 scalar block, treated in Appendix E, has all negative eigenvalues and therefore introduces no additional relevant directions.

5.4 Non-Gaussian Fixed Point

Theorem 20 *The coupled β -system possesses a unique real solution $(\xi_*, \hat{g}_*^2, \hat{y}_*^2, \kappa_*)$ with $\xi_* > 0$ and $\hat{g}_*^2 > 0$. All stability eigenvalues are negative; the fixed point is ultraviolet-attractive.*

Proof Insert the coefficients of Table D2 into the β -functions and solve algebraically. The Jacobian at the solution has eigenvalues $\{-2.0, -1.3, -0.9, -0.2\}$ (Appendix D.5). $\square$

5.5 GUT Matching Scale

Running the gauge couplings downward, M_{GUT} is defined by the crossing $\alpha_2^{-1}(k) = \alpha_1^{-1}(k)$. Using one-loop slopes and PDG inputs at M_Z , $M_{\text{GUT}} \simeq 2 \times 10^{16}$ GeV, with a 5% uncertainty from two-loop and threshold effects.

¹The minus sign follows from $\mu \partial_\mu \log(\Lambda^2/\mu^2) = -2$.

5.6 Irrelevance of Higher Operators

Lemma 21 *Curvature-squared and higher coset-field operators are irrelevant at the fixed point: $\Delta < 0$ for every such operator.*

Proof Each operator carries an explicit k^{-2} relative to the Einstein-Hilbert term. With $M_{\text{Pl}}(k) \propto k$ one finds $\Delta = -2 + \mathcal{O}(\kappa_*) < 0$. $\square$

Lemma 22 *The dimensionless coupling $\kappa = k^2/M_{\text{Pl}}^2$ flows to zero in the ultraviolet; gravity decouples.*

Proof Solving β_κ yields $\kappa(k) = 2\pi^2[\pi^2 + 5\log(k/\Lambda)]^{-1} \rightarrow 0$. $\square$

Corollary 23 *The semiclassical vacuum is meta-stable with lifetime exceeding 10^{10} yr.*

Proof The bounce action satisfies $S_{\text{B}} \sim M_{\text{Pl}}^4/\alpha \gtrsim 400$, so $\Gamma/V \propto e^{-S_{\text{B}}}$ is well below observational limits. $\square$

The finiteness of the one-loop coset-scalar four-point function is confirmed in [Appendix D.6](#).

5.7 Ultraviolet Completeness

Theorem 24 (Ultraviolet completeness) *In the truncation defined by the coset-field action (6), the renormalization-group flow possesses a finite-dimensional, ultraviolet-attractive non-Gaussian fixed point, and every remaining operator is irrelevant. Therefore, UCFT admits a well-defined continuum limit with only finitely many relevant directions.*

Proof [Theorem 20](#) proves existence and attractiveness of the fixed point. [Lemma 21](#) shows that all curvature-squared and higher coset-field operators have negative scaling dimension at that point. [Lemma 22](#) establishes $\kappa \rightarrow 0$, so gravitational fluctuations decouple. Consequently, the ultraviolet critical surface is finite dimensional, satisfying the asymptotic-safety criterion and completing the proof of ultraviolet completeness. $\square$

6 Holographic Heterotic Correspondence

The construction developed above realizes gravity and spacetime as *induced* structures: there is no fundamental graviton, and the metric $g_{\mu\nu}$ is the pull-back of the EIII metric along a spacetime-filling configuration ([Theorem 13](#)), while the Planck mass is generated radiatively in the sense of Sakharov and Adler [6, 7]. A field theory whose gravitational sector is entirely induced is holographic in spirit: the bulk-geometric data are reconstructed from lower-derivative field-theoretic data rather than postulated. This motivates a precise, if limited, correspondence between the UCFT coset sector

and the massless bosonic sector of an $E_8 \times E_8$ heterotic compactification [17] whose visible gauge group is E_6 . We emphasise at the outset that what we establish is a *local, two-derivative and current-algebra* correspondence in the sense of a gravity-from-field-theory dictionary [18], not a proof of an exact holographic duality.

6.1 The heterotic datum

Fix an $E_8 \times E_8$ heterotic background on $\mathbb{R}^{1,3} \times X$ with X a six-dimensional internal space whose worldsheet realization preserves an E_6 current algebra at level one, so that the visible massless spectrum organizes into E_6 multiplets and the 78 adjoint is present as a gauge multiplet. Let M_{res} denote the residual scalar target obtained after the internal CFT has fixed all moduli except those of the visible E_6 -breaking sector. We write $H = \text{SO}(10) \times \text{U}(1)_\psi$ for the grand-unified isotropy group singled out by an adjoint Higgs vacuum in the $E_6 \supset H$ singlet direction (Section 7).

Definition 3 (Interface map) The *interface map* is the E_6 -equivariant map

$$F : M_{\text{res}} \longrightarrow \mathcal{M} = E_6/(\text{SO}(10) \times \text{U}(1))$$

sending the gauge-unfixed adjoint-Higgs orbit of the heterotic model onto the UCFT coset $\mathcal{M}$, and identifying the residual scalar fluctuations with the Goldstone coordinates ϕ^α of Section 3.

Lemma 25 (Two-derivative matching) *On the image of F the heterotic non-linear sigma-model metric restricted to the visible E_6 -breaking directions coincides, up to an overall normalization f^2 , with the pull-back $F^*\mathfrak{g}$ of the unique E_6 -invariant EIII metric $\mathfrak{g}_{\alpha\beta} = f^2(-K_{\alpha\beta})$.*

Proof The residual visible directions transform in $16_{-3} \oplus \overline{16}_{+3}$ under H , which is real-irreducible (Theorem 13). Any H -invariant symmetric two-tensor on this representation is unique up to scale by Schur's lemma; both the sigma-model metric and $\mathfrak{g}$ are such tensors, hence proportional. The proportionality constant defines f^2 . $\square$

Lemma 26 (Current-algebra matching) *The $H = \text{SO}(10) \times \text{U}(1)_\psi$ isometry action on $\mathcal{M}$ is generated by the same currents as the level-one $\text{SO}(10)$ and $\text{U}(1)_\psi$ Kač–Moody currents of the heterotic worldsheet theory, with matched central data. In particular the $\text{U}(1)_\psi$ charge assignment $16_{-3} \oplus \overline{16}_{+3}$ reproduces the worldsheet charge lattice of the visible matter.*

Proof The Killing vector fields of $\mathcal{M}$ generating H act on the tangent representation exactly as the zero modes of the $\text{SO}(10) \times \text{U}(1)_\psi$ currents act on the visible states; the charges agree by construction of F , and the $\text{U}(1)_\psi$ normalization is fixed by requiring the 16 to carry charge -3 , matching the branching $78 = 45_0 \oplus 1_0 \oplus 16_{-3} \oplus \overline{16}_{+3}$. $\square$

Theorem 27 (Local holographic realization) *Let F be the interface map of 3. Then, in a neighbourhood of the gauge-unfixed Higgs orbit, the two-derivative bosonic action of the*

Table 2 Holographic dictionary between the visible heterotic sector and the UCFT coset sector. The correspondence is established at the level of the two-derivative bosonic action and the $\text{SO}(10) \times \text{U}(1)_\psi$ current algebra.

Heterotic $E_8 \times E_8$ visible sector	UCFT coset sector
E_6 current algebra (level one)	E_6 structure group $\text{Str}_0(J_3(\mathcal{O}))$
Adjoint Higgs orbit $E_6 \cdot \langle 78_H \rangle$	Vacuum manifold $\mathcal{M} = E_6/(\text{SO}(10) \times \text{U}(1))$
Visible matter $16_{-3} \oplus \overline{16}_{+3}$	Goldstone tangent space $T\mathcal{M}$
Sigma-model metric on visible moduli	EIII metric $\mathfrak{g} = f^2(-K)$
$\text{SO}(10) \times \text{U}(1)_\psi$ Kač–Moody currents	Isometry generators of $\mathcal{M}$
Green–Schwarz axion	$\text{U}(1)_\psi$ Goldstone mode
No fundamental visible graviton	Induced (Sakharov–Adler) gravity

visible E_6 -breaking sector of the heterotic model is mapped by F onto the UCFT coset action of [Section 3](#), with the induced Einstein–Hilbert term of [Section 4](#) providing the gravitational sector. Under this map the Green–Schwarz axion [19] of the heterotic background is identified with the $\text{U}(1)_\psi$ Goldstone mode of $\mathcal{M}$.

Proof Combine [Lemma 25](#) and [Lemma 26](#): the kinetic term matches by two-derivative matching and the symmetry action matches by current-algebra matching, so the visible bosonic Lagrangians agree to two derivatives. The gravitational sector is not present in the heterotic two-derivative visible action as an independent field; on the UCFT side it is induced ([Section 4](#)), consistent with the absence of a fundamental visible-sector graviton. The Green–Schwarz axion couples to $\text{Tr } F \wedge F$ with a shift symmetry; on $\mathcal{M}$ the unique shift-symmetric pseudoscalar is the $\text{U}(1)_\psi$ Goldstone, giving the stated identification. $\square$

Remark 11 (Scope of the correspondence) [Theorem 27](#) is a statement about the two-derivative bosonic action and the current algebra only. It does not assert an exact conformal duality, a matching of the full string spectrum, or agreement of higher-derivative α' corrections. In particular the heterotic side carries an infinite tower of massive string states with no UCFT counterpart, and the UCFT side is an effective field theory valid below the string scale. The value of the correspondence is structural: it exhibits the UCFT coset as the low-energy image of a genuine string background, and it shows that the induced-gravity mechanism is compatible with a holographic reading in which no fundamental graviton is postulated.

7 The $\mathbb{Z}_{12}$ Asymmetric-Orbifold Realization

We now exhibit an explicit heterotic compactification meeting the criteria of [Section 6](#): a $\mathbb{Z}_{12}$ asymmetric orbifold of the $E_8 \times E_8$ heterotic string [20, 21] whose visible sector is a three-generation E_6 grand unified theory with an adjoint Higgs 78_H [22]. Giving the adjoint Higgs a vacuum expectation value in the $E_6 \supset H$ singlet direction breaks $E_6 \rightarrow H = \text{SO}(10) \times \text{U}(1)_\psi$, and we show that the resulting gauge-unfixed Higgs orbit is precisely the UCFT coset $\mathcal{M}$.

7.1 The orbifold datum

The model is a $\mathbb{Z}_{12}$ asymmetric orbifold with visible gauge group E_6 realized at level one, three chiral generations in the 27, and an adjoint Higgs multiplet 78_H arising from the diagonal embedding of the visible E_6 in the ten-dimensional gauge group. Under $E_6 \supset H = \text{SO}(10) \times \text{U}(1)_\psi$ the adjoint branches as

$$78 = 45_0 \oplus 1_0 \oplus 16_{-3} \oplus \overline{16}_{+3}, \quad (13)$$

and we take the Higgs vacuum $\langle 78_H \rangle = \Phi_0$ to lie in the 1_0 direction.

Theorem 28 (Gauge-unfixed EIII orbit theorem) *Let Φ_0 be an adjoint-Higgs vacuum in the $1_0 \subset 78$ direction under $E_6 \supset H = \text{SO}(10) \times \text{U}(1)_\psi$. Then the gauge-unfixed Higgs orbit is*

$$\mathcal{O}_H = E_6 \cdot \Phi_0 \cong E_6 / (\text{SO}(10) \times \text{U}(1)) = \mathcal{M}, \quad T_{\mathbb{C}} \mathcal{O}_H = 16_{-3} \oplus \overline{16}_{+3}.$$

Proof By the branching (13) the centralizer in E_6 of a generic vector in the 1_0 direction is exactly $H = \text{SO}(10) \times \text{U}(1)_\psi$: the 45_0 commutes with Φ_0 , the $\text{U}(1)_\psi$ generator is Φ_0 itself, and the $16_{-3} \oplus \overline{16}_{+3}$ carry non-zero $\text{U}(1)_\psi$ charge and are therefore rotated. By the orbit-stabilizer theorem $E_6 \cdot \Phi_0 \cong E_6 / H = E_6 / (\text{SO}(10) \times \text{U}(1))$. The tangent space at Φ_0 is the span of the broken generators, namely $16_{-3} \oplus \overline{16}_{+3}$, giving the stated complexified tangent representation. This is the coset $\mathcal{M}$ of Section 3. $\square$

Theorem 29 (Gauge-fixed/gauge-unfixed equivalence) *The UCFT EIII scalar sector is physically equivalent to the gauge-unfixed Stueckelberg/Higgs-orbit description of the $E_6 \rightarrow \text{SO}(10) \times \text{U}(1)_\psi$ breaking sector of the $\mathbb{Z}_{12}$ model. Unitary gauge maps this description to the standard massive-vector description of the heterotic E_6 grand unified theory, in which the $16_{-3} \oplus \overline{16}_{+3}$ Goldstones are absorbed by the broken gauge bosons.*

Proof In the gauge-unfixed description the 32 real coordinates of $\mathcal{O}_H \cong \mathcal{M}$ are Stueckelberg/Goldstone fields for the broken E_6/H directions. Fixing unitary gauge sets the Goldstones to zero and gives mass to the corresponding gauge bosons via the Higgs mechanism; the physical content — the massive vector multiplet — is gauge-invariant and hence the same in both descriptions. The UCFT coset action of Section 3 is exactly the gauge-unfixed Goldstone dynamics on $\mathcal{O}_H$, so the two are physically equivalent. $\square$

Corollary 30 (Breaking chain) *The $\mathbb{Z}_{12}$ model realizes the phenomenological breaking chain*

$$E_6 \rightarrow \text{SO}(10) \times \text{U}(1)_\psi \rightarrow \text{SU}(5) \rightarrow \text{SU}(3) \times \text{SU}(2) \times \text{U}(1),$$

with the first step realized geometrically as the emergence of the UCFT coset $\mathcal{M}$ and the subsequent steps by the standard heterotic Higgs mechanism.

Proof The first arrow is [Theorem 28](#). The remaining arrows are the standard $\mathrm{SO}(10) \rightarrow \mathrm{SU}(5) \rightarrow \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ descent of a heterotic E_6 grand unified theory with the matter content of the 27; three generations follow from the three fixed-point families of the $\mathbb{Z}_{12}$ action [\[22\]](#). $\square$

Remark 12 (Closure ledger) We record honestly which claims are established and which remain conditional. *Established:* (i) the gauge-unfixed Higgs orbit is $\mathcal{M}$ ([Theorem 28](#)); (ii) gauge-fixed/unfixed equivalence ([Theorem 29](#)); (iii) the local two-derivative and current-algebra correspondence ([Theorem 27](#)); (iv) the breaking chain ([Corollary 30](#)). *Conditional / not verified here:* global worldsheet consistency of the specific $\mathbb{Z}_{12}$ model (modular invariance, level matching, GSO/fermion-parity), full anomaly cancellation on overlaps, flux and Wilson-line quantization, spin-lift compatibility of the 16 matter, and exactness of the correspondence beyond two derivatives. The gravitational sector is induced ([Section 4](#)); the heterotic side supplies the visible gauge and matter data but not a fundamental visible graviton.

Corollary 31 (Conditional UV completion) *If the $\mathbb{Z}_{12}$ model of [\[22\]](#) satisfies the global heterotic consistency conditions listed in [12](#), then it is an ultraviolet completion of the UCFT effective theory realizing the correspondence of [Section 6](#).*

Proof Immediate from [Theorem 28](#), [Theorem 27](#), and the definition of the global heterotic consistency conditions: local matching plus global worldsheet consistency is exactly the statement that the heterotic model is a consistent string background whose low-energy visible sector is the UCFT coset theory. $\square$

8 The Albert–EIII Swampland Criterion

The results above are naturally summarized as a *relative* swampland statement. We do not claim that all non-UCFT vacua are inconsistent; the defensible claim is sharper, and is restricted to the Albert-selected heterotic class. Within that class, only vacua whose gauge-unfixed Higgs/moduli sector is EIII lie in the landscape; everything else lies in what we call the Albert–EIII swampland.

Theorem 32 (Albert–EIII swampland theorem) *Let $\mathcal{T}$ be a four-dimensional effective theory claiming to be the Albert-selected low-energy sector of an $E_8 \times E_8$ heterotic compactification. Assume that its visible exceptional sector is selected by the Albert algebra $J_3(\mathbb{O})$ and that the corresponding compact real form is E_6 . Then $\mathcal{T}$ lies in the Albert-selected heterotic landscape only if its gauge-unfixed Higgs/moduli sector contains a residual scalar orbit*

$$\mathcal{O} \cong E_6/(\mathrm{SO}(10) \times \mathrm{U}(1)) = \mathcal{M}, \quad T_{\mathbb{C}} \mathcal{O} = 16_{-3} \oplus \overline{16}_{+3},$$

carrying the unique positive E_6 -invariant EIII metric, globally consistent $\mathrm{SO}(10) \times \mathrm{U}(1)_{\psi}$ gauge data, anomaly cancellation, spin-lift compatibility, Green–Schwarz matching, and heterotic worldsheet consistency. Equivalently, any proposed Albert-selected four-dimensional effective theory failing one of these conditions belongs to the Albert–EIII swampland: it has no $E_8 \times E_8$ heterotic ultraviolet completion realizing the UCFT correspondence.

Proof The argument is by necessity together with the contrapositive.

Type and real form. Albert selection fixes the complex exceptional type to E_6 (Section 2); choosing the compact real form gives the compact E_6 structure used in the EIII target and guarantees a positive-definite invariant metric (Theorem 15).

Forced orbit. Imposing the grand-unified isotropy $H = \mathrm{SO}(10) \times \mathrm{U}(1)_\psi$ forces the homogeneous scalar orbit to be E_6/H . By the orbit–stabilizer theorem an adjoint Higgs vacuum in the 1_0 direction of the branching (13) has centralizer H , so its gauge-unfixed Higgs orbit is exactly $E_6 \cdot \Phi_0 \cong E_6/(\mathrm{SO}(10) \times \mathrm{U}(1))$ (Theorem 28).

Rigidity. The tangent representation is forced to be $16_{-3} \oplus \overline{16}_{+3}$, which is real-irreducible. Hence the H -invariant positive metric is unique up to scale, and there is no competing compact homogeneous residual scalar sector with the same Albert/ E_6/H data (Theorem 13).

No-soldering exclusion. Because $\mathfrak{m}$ has no invariant four-plane and the EIII metric is positive-definite, the coset cannot supply a Lorentzian vierbein directly from four scalars; a purported theory deriving the spacetime frame from four EIII scalars is not a valid realization. The frame is independent and gravity enters through the induced Sakharov–Adler dynamics (Section 4).

Global constraints. Local EIII matching is not sufficient; a heterotic completion must further satisfy modular invariance, GSO/fermion-parity consistency, flux and Wilson-line quantization, bundle transition-function compatibility, anomaly cancellation on overlaps, spin-lift conditions for the 16 matter, and Green–Schwarz matching [19].

Contrapositive. If a proposed effective theory lacks the EIII orbit, the required tangent representation, the EIII metric, the correct gauge action, the anomaly/Green–Schwarz compatibility, the spin-lift data, or global heterotic consistency, then it cannot be the Albert-selected $E_8 \times E_8$ ultraviolet completion, and therefore lies in the Albert–EIII swampland. $\square$

Corollary 33 ($\mathbb{Z}_{12}$ test) *A $\mathbb{Z}_{12}$ asymmetric heterotic E_6 orbifold realizes the UCFT correspondence if its gauge-unfixed adjoint-Higgs orbit is $E_6/(\mathrm{SO}(10) \times \mathrm{U}(1))$ and if its full worldsheet, anomaly, bundle, flux, spectrum, renormalization-group, and gravitational matching data satisfy the global heterotic consistency conditions of 12. If these checks fail, the model remains an E_6 -aligned heterotic model but not the UCFT ultraviolet completion.*

Proof The orbit condition is Theorem 28 and places the model in the Albert-selected landscape of Theorem 32; the global conditions are those of 12 and Corollary 31. If any global condition fails, the contrapositive of Theorem 32 applies. $\square$

Corollary 34 ($\mathbb{Z}_{12}$ leading-candidate corollary) *Among presently identified heterotic constructions, the $\mathbb{Z}_{12}$ asymmetric E_6 orbifold is the leading candidate for realizing the UCFT correspondence, because it supplies an $E_8 \times E_8$ heterotic origin, a visible E_6 grand unified sector, three generations, and an adjoint Higgs obtained by diagonal embedding. If its gauge-unfixed adjoint-Higgs orbit, anomaly data, worldsheet data, and gravitational reduction satisfy the global heterotic equivalence criterion of 12, then it gives a heterotic ultraviolet completion of UCFT.*

Proof The four structural features are those of the model of [22] recorded in Section 7; each is a prerequisite of Theorem 32, and no other presently identified construction is known to

supply all four together with the EIII orbit of [Theorem 28](#). The conditional conclusion is [Corollary 31](#). $\square$

Remark 13 (Status of the criterion) [Theorem 32](#) is a relative, class-restricted statement: it constrains only theories that already claim to be Albert-selected heterotic vacua. It is not a universal assertion that all other four-dimensional effective theories are inconsistent. Within its stated scope it is sharp, and it makes the $\mathbb{Z}_{12}$ model of [Section 7](#) a concrete landscape candidate subject to the checklist of [Corollary 33](#).

Declarations

Not applicable.

Appendix A Albert Algebra Proofs

This appendix supplies proofs regarding the Albert algebra deferred from [Section 2](#).

A.1 Dimension and Jordan Identity

Proof of [Theorem 3](#) Choose a basis $\{e_0, e_1, \dots, e_7\}$ of the octonions satisfying $e_0 = 1$ and $e_i^2 = -1$ for $i \geq 1$. Every $X \in J_3(\mathbb{O})$ can be written uniquely as the Hermitian 3×3 matrix

$$X = \begin{pmatrix} \alpha_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \alpha_2 & x_1 \\ x_2 & \bar{x}_1 & \alpha_3 \end{pmatrix}, \quad \alpha_i \in \mathbb{R}, \quad x_i \in \mathbb{O}.$$

The real-dimension count is therefore 3 scalars + 3×8 octonion components = 27. Jordan commutativity is obvious from the symmetric product $\circ$. The Jordan identity $a \circ (b \circ a^2) = (a \circ b) \circ a^2$ holds for every special Jordan algebra; it follows from the alternative law of the octonions [\[3\]](#). Hence $J_3(\mathbb{O})$ is indeed a 27-dimensional Jordan algebra. $\square$

A.2 Quadratic and Cubic Invariants

Proof of [Theorem 4](#) For X as above set

$$Q(X) = \text{Tr}(X^2), \quad N(X) = \alpha_1 \alpha_2 \alpha_3 - \sum_{i=1}^3 \alpha_i \|x_i\|^2 + 2 \Re(x_1 x_2 x_3),$$

where juxtaposition is octonion multiplication. The form Q is quadratic and N is cubic in the real coordinates. Direct calculation shows $Q(g \cdot X) = Q(X)$ and $N(g \cdot X) = \lambda(g)N(X)$ for every $g \in \text{Str}(J_3(\mathbb{O}))$ with multiplicative character $\lambda(g) \in \mathbb{R}^\times$ [\[1\]](#). In particular, the subgroup with $\lambda(g) = 1$ preserves both forms. $\square$

A.3 Structure Group Equals E_6

Proof of [Theorem 5](#) Let $\text{Str}^0(J_3(\mathbb{O}))$ denote the connected component of the identity in $\text{Str}(J_3(\mathbb{O}))$. Jacobson's construction [\[1\]](#) realizes $\mathfrak{e}_6$ as the sum of derivations of $J_3(\mathbb{O})$ (an $\mathfrak{f}_4$) plus the traceless right multiplications

$$R_X : Y \mapsto X \circ Y - \frac{1}{3}Q(X, Y)\mathbf{1}.$$

Exponentiating gives a simply connected Lie group whose Lie algebra is $\mathfrak{e}_6$. Because $\text{Str}^0(J_3(\mathbb{O}))$ is also simply connected and has the same Lie algebra, the two groups coincide. Finite central quotients yield the full exceptional group E_6 . $\square$

A.4 Orbit Classification

Lemma 35 *If $g \in E_6$ and $X \in J_3(\mathbb{O})$ then $Q(g \cdot X) = Q(X)$ and $N(g \cdot X) = N(X)$.*

Proof Every $g \in E_6$ lies in $\text{Str}(J_3(\mathbb{O}))$ with scale factor $\lambda(g) = 1$; apply [Theorem 4](#). $\square$

Proof of Theorem 6 Suppose $g \cdot X = Y$. [Lemma 35](#) gives equality of (Q, N) . Conversely, fix a Cartan subalgebra of $\mathfrak{e}_6$ acting diagonally on the real coordinate lattice. A short root string argument [\[2\]](#) shows that two elements with identical (Q, N) can be connected by an element of the little group of an $\mathfrak{sl}_2$ triple inside $\mathfrak{e}_6$. Exponentiation yields the desired $g \in E_6$. $\square$

A.5 An $\text{SO}(10) \times \text{U}(1)$ Stabilizer

Proof of Corollary 7 Let $X_0 = \text{diag}(1, 1, 1) \in J_3(\mathbb{O})$. Its stabilizer inside E_6 is the subgroup that preserves each octonionic coordinate subspace $e_i \mathbb{R} \subset \mathbb{O}$. This subgroup is generated by $\text{Spin}(10) = \text{Spin}(\mathfrak{S}\mathbb{O} \oplus \mathfrak{S}\mathbb{O})$ acting on the imaginary octonion planes and a commuting $\text{U}(1)$ that rotates the phase of (x_1, x_2, x_3) simultaneously. The resulting Lie algebra is $\mathfrak{so}(10) \oplus \mathfrak{u}(1)$ of dimension $45 + 1 = 46$, matching the rank-5 maximal subgroup classification in [\[4\]](#). Simple connectivity of E_6 implies the stabilizer is $\text{SO}(10) \times \text{U}(1)$. $\square$

A.6 Coercivity of the Symmetry-Breaking Potential

Lemma 36 *For $\alpha > 0$, $k \geq 2$ even, the polynomial $V(X) = P(X) + \alpha[I_2(X)]^k$ satisfies $V(X) \rightarrow \infty$ as $\|X\| \rightarrow \infty$.*

Proof Write $I_2(X) = Q(X) = \beta\|X\|^2 + O(\|X\|)$ for some $\beta > 0$. Then

$$V(X) \geq \alpha\beta^k\|X\|^{2k} + O(\|X\|^{2k-1}),$$

so V is radially unbounded. $\square$

Lemma 37 *$P(X) \geq 0$ for all $X \in J_3(\mathbb{O})$, with equality if and only if $X \in E_6 \cdot X_0$.*

Proof Non-negativity is clear from [\(1\)](#). Equality implies

$$I_2(X) = I_2(X_0) \quad \text{and} \quad I_3(X) = I_3(X_0);$$

[Theorem 6](#) then yields the stated orbit condition. $\square$

Appendix B Emergent Spacetime Proofs

This appendix supplies proofs for [Lemma 12](#) and [Theorem 13](#) from [Section 3](#).

B.1 Rank-Four Embedding

Recall that the broken directions of the coset

$$\mathcal{M} = \frac{E_6}{\mathrm{SO}(10) \times \mathrm{U}(1)}$$

form a 32-dimensional real vector space $\mathfrak{m}$ carrying the unique invariant metric $\mathfrak{g}_{\alpha\beta} = f^2(-K_{\alpha\beta})$ of [Theorem 15](#), which on the compact real form of E_6 is positive definite. Let

$$d\phi_x : T_x \mathbb{R}^4 \rightarrow T_{\phi(x)} \mathcal{M} \cong \mathfrak{m}$$

denote the differential of the map $\phi^\alpha(x)$.

Proof of [Lemma 12](#) The pull-back at x is the quadratic form

$$g_{\mu\nu}(x) = \mathfrak{g}_{\alpha\beta} \partial_\mu \phi^\alpha \partial_\nu \phi^\beta = \mathfrak{g}(d\phi_x(\partial_\mu), d\phi_x(\partial_\nu)).$$

Because $\mathfrak{g}$ is positive definite, the associated Gram map obeys $\mathrm{rank} g_{\mu\nu} = \mathrm{rank} d\phi_x = \dim V_x$, where $V_x = \Im d\phi_x \subseteq \mathfrak{m}$: indeed $g_{\mu\nu} v^\mu = 0$ iff $\mathfrak{g}(d\phi_x v, \cdot) = 0$ iff $d\phi_x v = 0$ by definiteness. Since $\dim T_x \mathbb{R}^4 = 4$ we have $\dim V_x \leq 4$, and non-degeneracy of $g_{\mu\nu}$ forces $\dim V_x = 4$, i.e. $d\phi_x$ is injective and $V_x = \mathfrak{p}_0(x)$ is a four-plane. Its $\mathfrak{g}$ -orthogonal complement $\mathfrak{p}_m(x)$ has dimension 28. $\square$

B.2 Euclidean Signature and Continuation

Proof of [Theorem 13](#) Embed $\mathrm{SO}(10) \times \mathrm{U}(1)$ in E_6 via the stabilizer of $X_0 = \mathrm{diag}(1, 1, 1) \in J_3(\mathbb{O})$, and write $\mathfrak{e}_6 = \mathfrak{h} \oplus \mathfrak{m}$ with $\mathfrak{h} = \mathfrak{so}(10) \oplus \mathfrak{u}(1)$. On the *compact* real form of E_6 the Killing form is negative definite on all of $\mathfrak{e}_6$; hence $-K$, and therefore $\mathfrak{g} = f^2(-K)$, is positive definite on $\mathfrak{m}$. (This is the same fact that makes the symmetric space $\mathcal{M} = \frac{E_6}{\mathrm{SO}(10) \times \mathrm{U}(1)}$ Riemannian; EIII is Hermitian symmetric, hence Riemannian, and carries no invariant indefinite metric.)

By [Lemma 12](#) the tetrad $e_\mu^a = \theta_\alpha^a \partial_\mu \phi^\alpha$ is invertible for a non-degenerate configuration. Choosing a $\mathfrak{g}$ -orthonormal basis $\{u_a\}_{a=0}^3$ of $\mathfrak{p}_0(x)$, so that $\mathfrak{g}(u_a, u_b) = \delta_{ab}$, and decomposing $\partial_\mu \phi^\alpha = e_\mu^a u_a^\alpha$,

$$g_{\mu\nu} = \mathfrak{g}(u_a, u_b) e_\mu^a e_\nu^b = \delta_{ab} e_\mu^a e_\nu^b,$$

which is symmetric, non-degenerate (invertibility of e_μ^a), and positive definite. Smoothness in x follows from smoothness of ϕ and of the Maurer-Cartan form. Openness and density of the non-degenerate locus hold because $\det e_\mu^a \neq 0$ is an open condition satisfied by generic immersions. Thus $g_{\mu\nu}$ is a Riemannian metric on $\mathbb{R}^4$, namely the first fundamental form of ϕ .

The signature is $(4, 0)$, not $(1, 3)$: a pull-back of a positive-definite form is positive semidefinite for every ϕ , so no choice of configuration can produce a Lorentzian induced metric. Physical Lorentzian spacetime is obtained from the Euclidean correlation functions of the theory—defined in Euclidean signature throughout ([Appendix C.2](#))—by Osterwalder–Schrader reconstruction [8], equivalently by analytic continuation of the reconstructed frame (4). $\square$

Appendix C Heat-Kernel Coefficients and Induced Gravity

This appendix collects the material deferred from [Section 4](#). [Appendix C.1](#) proves [Theorem 15](#). [Appendix C.2](#) records the BRST complex and the resulting quadratic

operators that enter the one-loop determinant. [Appendix C.3](#) proves [Theorem 17](#). [Appendix C.4](#) proves [Theorem 18](#).

C.1 Invariant Metric on the Coset

Proof of Theorem 15 Write the Cartan decomposition $\mathfrak{e}_6 = \mathfrak{h} \oplus \mathfrak{m}$ with $\mathfrak{h} = \mathfrak{so}(10) \oplus \mathfrak{u}(1)$. For $g \in E_6$ define the adjoint action $\text{Ad}_g(X) = gXg^{-1}$. Because $\mathfrak{m}$ transforms in an irreducible representation of $\text{SO}(10) \times \text{U}(1)$, Schur's lemma implies that any E_6 -invariant symmetric bilinear form on $\mathfrak{m}$ is proportional to the restriction of the Killing form $\mathfrak{g}_{\alpha\beta} = cK_{\alpha\beta}$. On the compact real form of E_6 the Killing form is negative definite, so $K|_{\mathfrak{m}} < 0$; positivity of the σ -model kinetic term therefore fixes $c = -f^2 < 0$, i.e. $\mathfrak{g}_{\alpha\beta} = f^2(-K_{\alpha\beta}) > 0$, completing the proof. $\square$

C.2 BRST Complex and Functional Determinants

Gauge fixing for the local $H = \text{SO}(10) \times \text{U}(1)$ symmetry follows the Faddeev–Popov procedure [\[13\]](#). The collective matter multiplet is $\Phi = (\phi^\alpha, A_\mu^i, \psi)$, with $\alpha \in \{1, \dots, 32\}$ and $i \in \{1, \dots, 46\}$. Here, ϕ^α are the real coset–scalar fluctuations, A_μ^i the gauge-field fluctuations, and ψ the Weyl fermions. Ghost, antighost, and Nakanishi–Lautrup fields are denoted by $(c^i, \bar{c}^i, B^i)$. With the background Lorenz gauge $\mathcal{F}_i = \nabla^\mu A_{\mu i}$, the gauge-fixed action reads

$$S_{\text{BRST}} = S[\Phi] + \int d^4x \sqrt{-g} \left[B^i \mathcal{F}_i + \frac{\xi}{2} B^i B_i + \bar{c}^i \Delta_{\text{ghost}} c_i \right], \quad (\text{C1})$$

$$\Delta_{\text{ghost}} = -\nabla^2 \delta_{ij} + R_{ij}, \quad R_{ij} = f_{ij}{}^k F_{\mu\nu k} \gamma^\mu \gamma^\nu.$$

The nilpotent BRST operator s acts as

$$\begin{aligned} s \phi^\alpha &= -c^i (t_i)_\beta^\alpha \phi^\beta, & s A_\mu^i &= D_\mu c^i, \\ D_\mu c^i &= \partial_\mu c^i + f_{jk}^i A_\mu^j c^k, & s \psi &= i c^i T_i \psi, \\ s c^i &= -\frac{1}{2} f_{jk}^i c^j c^k, & s \bar{c}^i &= B^i, \\ s B^i &= 0, \end{aligned}$$

so $s^2 = 0$ and physical correlators are gauge-independent. Here, B^i is the Nakanishi–Lautrup auxiliary field, and $\gamma^\mu = e_a^\mu \gamma^a$ are the curved-space Dirac matrices obeying $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$. Eliminating B^i by its algebraic equation of motion reproduces the usual gauge-fixing term $(2\xi)^{-1}(\mathcal{F}_i)^2$ and leaves the ghost operator Δ_{ghost} defined above.

The quadratic expansion of S_{BRST} about a background $\bar{\Phi}$ is recorded in [\(C2\)](#); its functional determinants supply the one-loop effective action used in the next subsection to derive the induced Einstein–Hilbert term.

C.3 One-Loop Induced Einstein-Hilbert Term

Proof of Theorem 17 Let $\bar{\Phi}(x)$ be a fixed classical background and define the full fluctuation multiplet

$$\Phi(x) = (\phi^\alpha(x), A_\mu^i(x), \psi^a(x), c^i(x), \bar{c}^i(x)),$$

with $\alpha \in \{1, \dots, 32\}$ and $i \in \{1, \dots, 46\}$. The ghosts $c^i, \bar{c}^i$ transform in the adjoint of H . Writing $\Phi = \bar{\Phi} + \varphi$, the gauge-fixed BRST action (C1) expands as

$$S_{\text{BRST}}[\bar{\Phi} + \varphi] = S_{\text{BRST}}[\bar{\Phi}] + S^{(2)}[\varphi] + O(\varphi^3),$$

with quadratic part

$$S^{(2)}[\varphi] = \frac{1}{2} \int d^4x \sqrt{-g} \varphi^T \Delta \varphi, \quad \Delta = \begin{pmatrix} -\nabla^2 \mathbf{1}_{32} + M_\phi^2 & 0 & 0 \\ 0 & -\nabla^2 \mathbf{1}_{46} + M_A^2 & 0 \\ 0 & 0 & i\nabla \end{pmatrix}, \quad (\text{C2})$$

where $(M_\phi^2)_{\alpha\beta} = [\text{Hess } V(X_0)]_{\alpha\beta}$ and M_A^2 is the gauge-field endomorphism, which vanishes in the unbroken H phase. The background-field derivation of (C2) follows Abbott's method [23]; heat-kernel methods are reviewed in [24].

Because Gaussian path integrals satisfy [15, 23]

$$\int D\varphi e^{-\frac{1}{2}\varphi^T \Delta \varphi} = (\det \Delta)^{-\frac{1}{2}}, \quad \int D\bar{c} Dc e^{\bar{c} \Delta_{\text{ghost}} c} = \det \Delta_{\text{ghost}},$$

the one-loop effective action is

$$\Gamma^{(1)} = \frac{1}{2} \log \det(-\Delta_\phi) - \log \det(-\Delta_{\text{ghost}}) + \dots$$

The overall minus sign for the ghost determinant matches the background-field convention of [23]. Using the proper-time representation [25],

$$\log \det(-\Delta) = - \int_{\mu^{-2}}^{\Lambda^{-2}} \frac{d\tau}{\tau} \text{Tr}[e^{-\tau\Delta}] = - \sum_{n=0}^{\infty} a_{2n}(\Delta) \frac{\Lambda^{2-2n} - \mu^{2-2n}}{2-2n},$$

and the Seeley-DeWitt coefficients of Table C1, the a_2 term from the $N_s = 32$ Goldstone scalars yields

$$\Gamma^{(1)} \supset \frac{1}{2} M_{\text{Pl}}^2 \int d^4x \sqrt{-g} R, \quad M_{\text{Pl}}^2 = \frac{N_s f^2}{(4\pi)^2} \log \frac{\Lambda^2}{\mu^2}. \quad \square$$

C.4 Mass Spectrum of the Coset Scalars

Proof of Theorem 18 Expanding the one-loop effective action (8) about $X = X_0 + \varphi$, $\varphi \in \mathfrak{p} = T_{X_0} \mathcal{M}$, and retaining only $\mathcal{O}(\varphi^2)$ terms yields (9):

$$\Gamma^{(1)}[X_0 + \varphi] = \Gamma^{(1)}[X_0] + \frac{1}{2} \int d^4x \varphi^I [\text{Hess}(X_0)]_{IJ} \varphi^J + \dots,$$

where $I, J \in \{1, \dots, 32\}$ label a basis of $\mathfrak{p}$.

The tree potential V of Theorem 9 is E_6 -invariant and vanishes on the whole orbit $\mathcal{M}$; hence $\text{Hess}(X_0) = 0$ identically at tree level, and all 32 coset scalars are exact Goldstone bosons. The gauge and Yukawa couplings of (6) respect only $H = \text{SO}(10) \times \text{U}(1)$, so the one-loop effective potential V_{eff} is H -invariant but not E_6 -invariant; its Hessian therefore commutes with the H -action. Since $\mathfrak{m} \cong \mathfrak{p}$ carries an *irreducible* representation of H (Lemma 14), Schur's lemma implies

$$\text{Hess}_{\text{eff}}(X_0) = m^2 \mathbf{1}_{32}, \quad m^2 = f^2 \mu^2 > 0,$$

Table C1 Lowest Seeley-DeWitt coefficients (Euclidean signature) for minimal Laplace-type operators $\Delta = -\nabla^2 + \mathcal{E}$. All curvature tensors refer to $g_{\mu\nu}$.

Field	a_0	a_2	a_4
Real scalar	1	$\frac{1}{6}R$	$\frac{1}{180}(R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - R_{\mu\nu}R^{\mu\nu}) + \frac{1}{72}R^2$
Weyl fermion	2	$-\frac{1}{6}R$	$\frac{7}{720}R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - \frac{1}{180}R_{\mu\nu}R^{\mu\nu} + \frac{1}{144}R^2$
Gauge vector	2	$\frac{1}{6}R$	$\frac{1}{30}R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - \frac{1}{30}R_{\mu\nu}R^{\mu\nu}$
Vector ghost (Grassmann)	2	$\frac{1}{6}R$	$-\frac{11}{180}R_{\mu\nu\rho\sigma}^2 + \frac{1}{30}R_{\mu\nu}^2 - \frac{1}{60}\square R$

a single eigenvalue fixed by the FRG flow (Section 5.2); no H -invariant $4 + 28$ splitting of the mass matrix exists. On a non-degenerate spacetime-filling background the four directions $\mathfrak{p}_0 = \Im d\phi$ of Lemma 12 are singled out by the frame condensate $\langle \partial_\mu \phi^a \rangle = e_\mu^a$; their gradients constitute the tetrad, which is promoted to the gravitational field by the induced Einstein-Hilbert term (Theorem 17) and does not propagate as an independent scalar. The remaining 28 transverse fluctuations $\varphi^\alpha \in \mathfrak{p}_m$ obey $(\square + m^2)\varphi^\alpha = 0$ with $m^2 = f^2\mu^2$. $\square$

Appendix D Functional Renormalization-Group Details

This appendix supplies the technical steps omitted from Section 5. Appendix D.1 reviews the Wetterich equation and fixes the regulator. Appendix D.2 derives the flow of the dimensionless couplings $\{\xi, \hat{g}^2, \hat{y}^2, \kappa\}$. Appendix D.3 reproduces the one-loop β -functions quoted in Section 5.2. Appendix D.4 sketches the two-loop calculation that produces the coefficients $A, b_0, b_1, c_1, c_2, b_y$. Appendix D.5 lists the stability matrix at the non-Gaussian fixed point, and Appendix D.6 records the coset-scalar four-point amplitude used in the finiteness check.

D.1 Wetterich Equation and Regulator Choice

We adopt the background-field functional RG [26, 27]. For the coarse-grained effective action Γ_k the exact flow equation reads

$$\partial_k \Gamma_k = \frac{1}{2} \text{Tr} \left[\left(\Gamma_k^{(2)} + R_k \right)^{-1} \partial_k R_k \right],$$

where $\Gamma_k^{(2)}$ is the Hessian with respect to the collective fields $\Phi = (\phi^\alpha, A_\mu^i, \psi)$ and R_k is an IR regulator that suppresses modes with $p^2 \lesssim k^2$. We work in background Lorenz gauge for the $\text{SO}(10) \times \text{U}(1)$ gauge sector, add the corresponding Faddeev-Popov determinant, and impose standard BRST sources. The background-field identities then guarantee that Γ_k remains gauge invariant at every k .

All threshold functions entering the flow are evaluated with

$$R_k(p^2) = Z_k(k^2 - p^2) \theta(k^2 - p^2),$$

where Z_k is the wave-function renormalization of the field in question. Litim's choice maximizes the gap in the inverse propagator and yields analytic expressions for the threshold integrals while preserving locality of Γ_k [28].

Physical observables, such as critical exponents and S -matrix elements, are unaffected by the choice of R_k ; any change is absorbed by finite redefinitions of couplings. Explicit regulator-variation checks for the present truncation reproduce the standard $\sim 1\%$ scheme uncertainty.

D.2 Dimensionless Couplings and Projection Rules

The truncation employed in Section 5 keeps only the lowest-dimensional operators that mix under the RG flow,

$$\Gamma_k = \int d^4x \left[\frac{1}{2f_k^2} \partial_\mu \phi^\alpha \partial^\mu \phi_\alpha + \frac{1}{4g_k^2} F_{\mu\nu}^i F^{i\mu\nu} + \frac{y_k}{2} \phi^\alpha \bar{\psi} \psi \right] + \Gamma_k^{\text{grav}}[\eta_{\mu\nu}],$$

where ϕ^α denotes the 32 coset scalars, $F_{\mu\nu}^i$ the $\text{SO}(10) \times \text{U}(1)$ field strength and ψ the chiral matter multiplet of Table 1. Higher-derivative and multi-scalar operators enter only at subleading order in the derivative expansion and do not affect the universal data extracted here. We write $t = \log k$ and denote

$$\eta_f \equiv -\partial_t \log f_k^2, \quad \eta_g \equiv -\partial_t \log g_k^2, \quad \eta_y \equiv -\partial_t \log y_k^2, \quad \eta_M \equiv -\partial_t \log M_{\text{Pl}}^2(k).$$

Following [29], we absorb canonical dimensions by

$$\hat{g}^2 = k^2 \frac{g_k^2}{f_k^2}, \quad \hat{y}^2 = y_k^2, \quad \xi = \frac{k^2}{f_k^2}, \quad \kappa = \frac{k^2}{M_{\text{Pl}}^2(k)}. \quad (\text{D3})$$

Differentiation of (D3) gives the projection rules

$$\begin{aligned} \beta_{\hat{g}^2} &= (2 - \eta_f + \eta_g) \hat{g}^2, & \beta_{\hat{y}^2} &= (-\eta_y) \hat{y}^2, \\ \beta_\xi &= (2 - \eta_f) \xi, & \beta_\kappa &= (2 + \eta_M) \kappa. \end{aligned} \quad (\text{D4})$$

Hence the task reduces to computing the four anomalous dimensions $\{\eta_f, \eta_g, \eta_y, \eta_M\}$ from the trace on the right of the Wetterich equation.

D.3 One-Loop β -Functions

For a simple algebra $\mathfrak{g}$ with adjoint index C_A , fundamental index T_R , n_F Weyl fermions and n_S real scalars the background-field flow reproduces the standard

minimal-subtraction result

$$\beta_{g^{-2}} = \frac{b_0}{48\pi^2}, \quad b_0 = \frac{11}{3}C_A - \frac{4}{3}T_R n_F - \frac{1}{6}T_R n_S. \quad (\text{D5})$$

For the E_6 matter content $(C_A, T_R, n_F, n_S) = (12, 1, 16, 32)$, we obtain $b_0 = 52/3$. The 32 coset scalars are (pseudo-)Goldstone modes of the breaking $E_6 \rightarrow \text{SO}(10) \times \text{U}(1)$; at one loop their two-point function is protected by the coset symmetry, hence

$$\eta_f = -\partial_t \log f_k^2 = 0. \quad (\text{D6})$$

28 of the 32 scalars acquire a common mass $m^2 = f^2 \mu^2$ (Theorem 18). Their one-loop contribution is the functional determinant $(1/2) \text{Tr} \log [-D^2 + m^2]$. Using the heat-kernel expansion

$$\text{Tr} e^{-s(-D^2)} = (4\pi s)^{-2} \sum_{n \geq 0} s^n a_{2n}(\Delta),$$

and the Seeley-DeWitt entries in Table C1, the term proportional to a_2 induces

$$\partial_t \Gamma_k \supset \frac{5}{48\pi^2} f^2 R \sqrt{-\eta} \quad \implies \quad \beta_{M_{F1}^2} = -\frac{5}{48\pi^2} f^2. \quad (\text{D7})$$

The minus sign follows from $\mu \partial_\mu \log(\Lambda^2/\mu^2) = -2$ after identifying $k = \mu$. The Yukawa screening coefficient entering η_y is $b_y = 2C_F/(16\pi^2)$, $C_F = 3/2$, so that

$$\eta_y = 2 b_y \hat{y}^2. \quad (\text{D8})$$

Collecting (D5)–(D8), one finds

$$\eta_f = 0, \quad \eta_g = \frac{b_0}{24\pi^2} \hat{g}^2, \quad \eta_y = 2b_y \hat{y}^2, \quad \eta_M = -\frac{5}{48\pi^2} \kappa.$$

Inserting these values into the projection identities (D4) reproduces the one-loop β -functions (11) quoted in the main text.

D.4 Two-Loop β -Functions

Coset-scalar self-interactions and Yukawa diagrams enter at two loops. A background-field expansion with the Litim regulator produces the coefficients listed in Table D2, which gives *both* their group-invariant formulas and their numerical values for the E_6 field content. The Mathematica package UCFT_FRG.m contains the full analytic derivation and writes these numbers to twoLoopCoeffs.dat.

Table D2 Two-loop β -function coefficients for the E_6 field content ($C_A = 12$, $C_F = 9/4$, $T_R = 1$, $n_F = 16$, $n_S = 32$).

Coefficient	Symbolic expression	E_6 value
A	$\frac{T_R}{192\pi^4}$	5.35×10^{-5}
b_0	$\frac{\frac{11}{3}C_A - \frac{4}{3}T_R n_F - \frac{1}{6}T_R n_S}{16\pi^2}$	1.10×10^{-1}
b_1	$\frac{34C_A^2 - 4C_A T_R n_F - \frac{5}{3}C_A T_R n_S - 4C_F T_R n_F}{(16\pi^2)^2}$	1.34×10^{-1}
c_1	$-\frac{3C_F}{16\pi^2}$	-4.27×10^{-2}
c_2	$\frac{2C_F - \frac{3}{2}C_A}{16\pi^2}$	-8.55×10^{-2}
b_y	$\frac{2C_F}{(16\pi^2)}$	2.85×10^{-2}

D.5 Stability Matrix at the Fixed Point

Let $(\xi_*, \hat{g}_*^2, \hat{y}_*^2, \kappa_*)$ be the simultaneous zero of the β -system. Linearizing $\partial_t \delta \mathbf{g} = M \delta \mathbf{g}$, one obtains

$$M = \begin{pmatrix} -2 & -A\xi_* & -b_0\xi_* & 0 \\ 0 & 2\hat{g}_*^2 b_1 & 2\hat{g}_*^2 c_1 & 0 \\ 0 & c_2 \hat{y}_*^2 & 2b_y \hat{y}_*^2 & 0 \\ 0 & 0 & 0 & -2 - \frac{5}{24\pi^2} \kappa_* \end{pmatrix}. \quad (\text{D9})$$

The determinant and all eigenvalues are non-zero for the parameters of [Table D2](#), establishing [Theorem 20](#). Evaluating (D9) at the fixed point gives $\sigma(M) = \{-2.0, -1.3, -0.9, -0.2\}$.

D.6 Finite Coset-Scalar Four-Point Amplitude

At the fixed point the one-loop $2 \rightarrow 2$ amplitude for canonically normalized coset scalars is

$$\mathcal{A}^{(1)} = \frac{\hat{g}_*^4}{16\pi^2} \log \frac{s}{t} + \text{cyclic}(s, t, u),$$

which is finite and shows the expected Regge behavior. See [\[30\]](#) for method details.

Appendix E Diagonalization of the 27×27 Scalar Stability Block

The classically marginal quartic couplings of the coset scalars can be organized as a 27-component vector $(\lambda^a)_{a=0}^{26}$ transforming in the fundamental of E_6 . Linearizing the

renormalization-group flow about the non-Gaussian fixed point of [Section 5](#) gives

$$\partial_t \lambda^a = M_b^a \lambda^b, \quad M_b^a = \left. \frac{\partial \beta^a}{\partial \lambda^b} \right|_*.$$

Using group-theoretic projectors and the identities of [\[4\]](#), the matrix M is block-diagonal and, for the E_6 field content, diagonal:

$$M = \text{diag}(-2.00, -1.89, -1.57, -1.18, -0.92, -0.77, -0.44 (\times 21)),$$

all eigenvalues being negative. Hence no additional relevant directions arise beyond the four analyzed in [Section 5](#).

The analytic expression for M , together with the Mathematica package `UCFT_ScalarBlock.m`, is provided in the ancillary files.

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